Why? Because Einstein had a unique capacity to *imagine* how the world might be constructed, to ‘see’ it in his mind. The equations, for him, came afterwards; they were the language with which to make concrete his visions of reality. For Einstein, the theory of general relativity is not a collection of equations: it is a mental image of the world arduously translated into equations.

The idea behind the theory is that spacetime curves. If spacetime had only two dimensions, and we lived on a sort of plane, it would be easy to imagine what it means to say that ‘physical space curves’. It would mean that the physical space in which we live is not like a flat table but resembles instead a surface with mountains and valleys. But the world we inhabit does not have only two dimensions, it has three. Four, in fact, when time is included. To imagine a curved space in four dimensions is more complicated, because in our habitual perception we do not have an intuition of a ‘larger space’ within which spacetime can curve. But Einstein’s imagination had no difficulty in intuiting the cosmic mollusc in which we are immersed, which can be squashed, stretched and twisted – and that constitutes the spacetime around us. It is thanks to this visionary clarity that Einstein managed to be the first to write the theory.

In the end, a degree of tension between Hilbert and Einstein did develop. A few days before Einstein made his successful equation public, Hilbert had sent an article to a periodical which shows just how close he had come to the same solution – and even today historians of science are faced with doubts when trying to evaluate the respective contributions of these two giants. At some point their relations cooled, and Einstein feared that Hilbert, more senior and powerful than him, would seek to attribute to himself too much of the merit for the construction of the theory. But Hilbert never claimed to be the first to discover general relativity – and in a world such as that of science, where often, too frequently, disputes over precedence become poisonous – the two gave a truly wonderful example of wisdom, clearing the field of all negative tension.

Einstein writes a marvellous letter to Hilbert, summarizing the profound sense of the shared course they had taken:

> There was a moment in which something like an irritation came between us, the origin of which I no longer want to analyse. I have fought against the bitterness which it provoked in me, and have succeeded completely in doing so. I again think of you with unclouded friendship, and I ask you to do the same for me. It is really a pity if companions such as we are, who have managed to forge a path aside from the pettiness of this world, could find anything other than joy in each other’s company.[3]

The cosmos